

EXPERIMENT - 1

DATE OF EXP : 13-01-26

AIM : To study and verify the truth table of logic gates.

CODE :

```
#include <stdio.h>

int main() {
    int a, b;

    printf("A B | AND OR NOR XOR XNOR\n");
    printf("-----\n");

    for (a = 0; a <= 1; a++) {
        for (b = 0; b <= 1; b++) {

            int AND = a & b;
            int OR  = a | b;
            int NOR = ~(a | b);
            int XOR = a ^ b;
            int XNOR = ~(a ^ b);
            //int XOR = (a & !b) | (!a & b);
            // int XNOR = (a & b) | (!a & !b);

            printf("%d %d | %d %d %d %d %d\n",
                a, b, AND, OR, NOR, XOR, XNOR);
        }
    }

    return 0;
}
```

OUTPUT :

A	B	AND	OR	NOR	XOR	XNOR
0	0	0	0	1	0	1
0	1	0	1	0	1	0
1	0	0	1	0	1	0
1	1	1	1	0	0	1